

Project Title

Travel Package Booking System

Project Report submitted for the partial fulfilment of grade for the course

Database Systems Engineering and Distributed Backend Development (25CS1302E)
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ABSTRACT

The Travel Package Booking System is an efficient database-driven application designed to let users search, book, and manage travel packages while ensuring accurate, consistent, and reliable storage of every transaction. In modern travel and tourism platforms, a well-designed database is essential for handling large volumes of user, package, booking, and payment data without redundancy, inconsistency, or loss of information. This project models users, travel packages, bookings, and payments as related entities connected through primary and foreign keys, and stores them in a normalized relational schema that removes duplicate data and enforces referential integrity across tables. The system uses core DBMS concepts such as normalization (up to 3NF), indexing on frequently searched fields like destination and travel date, and optimized SQL queries to retrieve package and booking information quickly and efficiently. ACID-compliant transactions ensure that a booking and its corresponding payment are committed together, preventing partial updates, double-bookings, and data anomalies even when multiple users book simultaneously. Concurrency control mechanisms such as row-level locking manage simultaneous access to the same package record, while well-designed indexes and query plans keep search and booking operations responsive as the dataset grows. The system can handle multiple users, packages, and bookings at once, and determines seat or slot availability accurately before confirming a transaction. By ensuring data consistency, eliminating redundancy, and enabling fast and reliable retrieval, the proposed solution improves the overall efficiency, trustworthiness, and scalability of the booking platform. The project demonstrates the practical application of database design principles and DBMS concepts in solving real-world data management problems for the travel and tourism domain. Overall, the Travel Package Booking System provides a simple, robust, and efficient approach to managing travel bookings and helps travel service providers achieve faster, more accurate, and more dependable package transactions.

Signature of the Guide